

Open convex domains have the same bidual projection bound

A candidate full solution to Question 1.5 of arXiv:1703.08384

Automatic Mathematics Project — run `fa_banach_001`, lane 3

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Status. Candidate full solution, likely valid, pending specialist review. The answer to Question 1.5 is affirmative, with exactly the Banach–Mazur bound from Theorem 1.1 of the source paper.

Abstract

Let E be a finite-dimensional normed space and let $\Omega \subset E$ be nonempty, open, and convex. We prove that $\mathcal{F}(\Omega)$ is complemented in its bidual by a projection of norm at most $d_{BM}(E, \ell_2^{\dim E})$. The argument first handles the Euclidean case: the nearest-point map onto $\bar{\Omega}$ is a nonexpansive retraction, so its linearization transfers the norm-one bidual projection for the whole Euclidean space to the domain. Since Ω is dense in its closure, their Lipschitz-free spaces are canonically isometric. A Banach–Mazur change of norm then gives the asserted sharp inherited bound.

1 Source question and exact answer

Cúth, Kalenda, and Kaplický prove that if E is a real normed space of finite dimension $d \geq 2$, then there is a projection

$$Q_E : \mathcal{F}(E)^{**} \longrightarrow \mathcal{F}(E), \quad \|Q_E\| \leq d_{BM}(E, \ell_2^d).$$

Their Question 1.5 asks whether the same statement holds for $\mathcal{F}(\Omega)$ when Ω is a nonempty open convex subset of E [1]. The source excerpt is reproduced in Figure 1.

All Lipschitz-free spaces below are over real scalars. If o is the distinguished point of a metric space M , we write $\mathcal{F}(M, o)$ when the basepoint needs to be visible. Let $\kappa_X : X \rightarrow X^{**}$ denote the canonical embedding.

Theorem 1.1 (Affirmative answer to Question 1.5). *Let $(E, \|\cdot\|_E)$ be a real normed space of finite dimension $d \geq 2$, let $\Omega \subset E$ be nonempty, open, and convex, and choose any $o \in \Omega$ as basepoint. There is a bounded linear map*

$$P_\Omega : \mathcal{F}(\Omega, o)^{**} \longrightarrow \mathcal{F}(\Omega, o)$$

such that

$$P_\Omega \kappa_{\mathcal{F}(\Omega, o)} = \text{id}_{\mathcal{F}(\Omega, o)} \quad \text{and} \quad \|P_\Omega\| \leq d_{BM}(E, \ell_2^d).$$

Thus the canonical copy of $\mathcal{F}(\Omega, o)$ is complemented in its bidual with exactly the estimate in Theorem 1.1 of the source paper.

$\dim E \geq 2$ (at least in some norm).

An easy consequence is the following statement.

Corollary 1.2. *Let E be a finite-dimensional normed space and let $M \subset E$ be a subset whose closure has nonempty interior in E . Then $\mathcal{F}(M)$ is complemented in $\mathcal{F}(M)^{**}$.*

Proof. By [19] Corollary 3.5], $\mathcal{F}(M)$ is isomorphic to $\mathcal{F}(E)$, so the statement follows immediately from Theorem 1.1. $\square$

Another consequence of the main result is the following (the second assertion follows from the John theorem [18]).

Corollary 1.3.

- *Let E be a finite-dimensional Euclidean space. Then $\mathcal{F}(E)$ is 1-complemented in its bidual.*
- *Let E be a normed space of finite dimension $d \geq 2$. Then $\mathcal{F}(E)$ is $\sqrt{d}$ -complemented in its bidual.*

Note that for non-Euclidean spaces the estimate of the norm of a projection depends on dimension. The following question seems to be open.

Question 1.4. *Is there a constant $C \geq 1$ such that $\mathcal{F}(E)$ is C -complemented in $\mathcal{F}(E)^{**}$ for any finite-dimensional normed space E ?*

Note that the isometric representation of $\mathcal{F}(\Omega)$ given in [4] works for any nonempty open convex subset Ω of a finite-dimensional normed space E . However, the estimate for the projection given in Theorem 1.1 is proved only in the case $\Omega = E$. Hence, the following question seems to be open.

Question 1.5. *Does the statement of Theorem 1.1 hold also for $\mathcal{F}(\Omega)$, where Ω is a nonempty open convex subset of a finite-dimensional normed space?*

The original motivation for our investigation was, as mentioned above, the problem of complementability of $\mathcal{F}(\ell^1)$ in the bidual. We have not answered this question, but in view of the main results the following problem seems to be natural.

Question 1.6. *Is $\mathcal{F}(\ell^2)$ complemented in $\mathcal{F}(\ell^2)^{**}$? Is it 1-complemented?*

Figure 1: Corollary 1.3 and Questions 1.4–1.6 on page 2 of the source paper. Question 1.5 is the target of this packet.

2 Proof intuition

There are only two ingredients. First put a Euclidean norm on the ambient vector space. For a closed convex subset C of a finite-dimensional Hilbert space H , the nearest-point map $r : H \rightarrow C$ is 1-Lipschitz and fixes C . The universal property of Lipschitz-free spaces turns r into a contractive linear retraction $\mathcal{F}(H) \rightarrow \mathcal{F}(C)$. Sandwiching the norm-one projection $\mathcal{F}(H)^{**} \rightarrow \mathcal{F}(H)$ from Corollary 1.3 of the source paper between this retraction and the bidual of the inclusion yields a norm-one projection onto $\mathcal{F}(C)$. Taking $C = \bar{\Omega}$ costs nothing because a metric space and its completion have the same Lipschitz-free space.

Second, choose an isomorphism from E to ℓ_2^d that realizes the Banach–Mazur distance. Its affine restriction to Ω linearizes to an isomorphism of free spaces. Conjugating the Euclidean projection by this isomorphism multiplies its norm by at most $\|T\|\|T^{-1}\| = d_{BM}(E, \ell_2^d)$.

3 The Euclidean-domain step

We record the elementary functorial facts so that the norm estimate is fully auditable.

Lemma 3.1 (Passing to the completion). *If a pointed metric space (M, o) is dense in a pointed metric space (C, o) , then the inclusion induces an onto linear isometry*

$$J : \mathcal{F}(M, o) \longrightarrow \mathcal{F}(C, o).$$

Proof. The linearization J of the inclusion is an isometry. Indeed, every real-valued Lipschitz function on M has a unique extension to C with the same Lipschitz constant, so the dual formula for the free-space norm gives norm equality on the span of the evaluation vectors and hence on its closure. The range of J is closed. If $x \in C$ and $x_n \in M$ tends to x , then

$$\|\delta(x_n) - \delta(x)\|_{\mathcal{F}(C)} = d(x_n, x) \longrightarrow 0.$$

Thus the range contains every $\delta(x)$, $x \in C$, and is therefore all of $\mathcal{F}(C, o)$. $\square$

Lemma 3.2 (A retract inherits the bidual projection). *Let (C, o) be a pointed metric subspace of (M, o) . Suppose that there is a 1-Lipschitz retraction $r : M \rightarrow C$. If $Q_M : \mathcal{F}(M, o)^{**} \rightarrow \mathcal{F}(M, o)$ satisfies $Q_M \kappa_{\mathcal{F}(M, o)} = \text{id}$ and $\|Q_M\| \leq \lambda$, then there is a map $Q_C : \mathcal{F}(C, o)^{**} \rightarrow \mathcal{F}(C, o)$ satisfying $Q_C \kappa_{\mathcal{F}(C, o)} = \text{id}$ and $\|Q_C\| \leq \lambda$.*

Proof. Let $I : \mathcal{F}(C, o) \rightarrow \mathcal{F}(M, o)$ be induced by inclusion and let $R : \mathcal{F}(M, o) \rightarrow \mathcal{F}(C, o)$ be induced by r . McShane extension shows that I is an isometry; the universal property gives $\|R\| \leq 1$ and $RI = \text{id}$. Set

$$Q_C = R Q_M I^{**}.$$

Then $\|Q_C\| \leq \lambda$. Naturality of the canonical embeddings gives, for every $u \in \mathcal{F}(C, o)$,

$$Q_C \kappa_{\mathcal{F}(C, o)} u = R Q_M \kappa_{\mathcal{F}(M, o)} I u = R I u = u.$$

$\square$

Lemma 3.3 (Euclidean convex domains). *Let H be a finite-dimensional Euclidean space and let $\Omega \subset H$ be nonempty, open, and convex. For every basepoint $o \in \Omega$, the space $\mathcal{F}(\Omega, o)$ is 1-complemented in its bidual.*

Proof. Put $C = \overline{\Omega}$. This is a nonempty closed convex subset of H . Its metric projection

$$r(x) = \operatorname{argmin}_{c \in C} \|x - c\|_2$$

is well-defined, fixes C , and is nonexpansive. For completeness, if $p = r(x)$ and $q = r(y)$, the variational inequalities for the two minimizers give

$$\langle x - p, q - p \rangle \leq 0, \quad \langle y - q, p - q \rangle \leq 0.$$

Adding them yields

$$\|p - q\|_2^2 \leq \langle p - q, x - y \rangle \leq \|p - q\|_2 \|x - y\|_2,$$

so $\|r(x) - r(y)\|_2 \leq \|x - y\|_2$. Also $r(o) = o$.

Corollary 1.3 of the source paper supplies a map $Q_H : \mathcal{F}(H, o)^{**} \rightarrow \mathcal{F}(H, o)$ of norm 1 that is the identity on the canonical copy of $\mathcal{F}(H, o)$ [1]. Lemma 3.2 applied to $C \subset H$ and r gives a norm-one bidual projection for $\mathcal{F}(C, o)$. Finally, Ω is dense in C , so Lemma 3.1 identifies $\mathcal{F}(\Omega, o)$ isometrically with $\mathcal{F}(C, o)$. Transporting the projection through this isometry completes the proof. $\square$

4 Banach–Mazur transport and proof of the theorem

Proof of Theorem 1.1. The finite-dimensional Banach–Mazur infimum is attained. Choose a linear isomorphism

$$T : E \longrightarrow \ell_2^d, \quad \|T\| \|T^{-1}\| = d_{BM}(E, \ell_2^d).$$

To keep basepoints explicit, use the affine map

$$A : \Omega \longrightarrow \Omega' := T(\Omega - o), \quad A(x) = T(x - o).$$

It sends o to 0, and Ω' is again nonempty, open, and convex. Its linearization is a Banach-space isomorphism

$$U : \mathcal{F}(\Omega, o) \longrightarrow \mathcal{F}(\Omega', 0), \quad U\delta(x) = \delta(Ax),$$

with

$$\|U\| \leq \|T\|, \quad \|U^{-1}\| \leq \|T^{-1}\|.$$

Lemma 3.3 gives a norm-one map $Q_2 : \mathcal{F}(\Omega', 0)^{**} \rightarrow \mathcal{F}(\Omega', 0)$ that is the identity on the canonical copy. Define

$$P_\Omega = U^{-1}Q_2U^{**}.$$

Then

$$\|P_\Omega\| \leq \|U^{-1}\| \|Q_2\| \|U^{**}\| \leq \|T^{-1}\| \|T\| = d_{BM}(E, \ell_2^d).$$

For $u \in \mathcal{F}(\Omega, o)$, naturality gives

$$P_\Omega \kappa_{\mathcal{F}(\Omega, o)} u = U^{-1} Q_2 \kappa_{\mathcal{F}(\Omega', 0)} U u = U^{-1} U u = u.$$

This is the required projection. □

Remark 4.1 (The one-dimensional case). *The source theorem and Question 1.5 are framed around $d \geq 2$. The same argument also covers $d = 1$: the source paper recalls that $\mathcal{F}(\mathbb{R})$ is isometric to $L^1(\mathbb{R})$ and is 1-complemented in its bidual, while closed intervals are nonexpansive retracts of $\mathbb{R}$.*

5 Verification notes and scope

The proof was audited at the following possible failure points.

1. **Open versus closed domain.** No projection onto the open set is claimed. The nearest-point map lands in $\overline{\Omega}$, and Lemma 3.1 proves that this has exactly the same free space as Ω .
2. **Metric projection.** Nonexpansiveness is used only after moving to a Euclidean norm. It need not hold for the original norm.
3. **Basepoint.** The Euclidean retraction fixes every point of the closed convex set, in particular the chosen basepoint. The affine map A sends that basepoint to zero before linearization.
4. **Norm accounting.** The Euclidean stage has norm one. The only loss is conjugation by U , bounded by $\|T\| \|T^{-1}\| = d_{BM}(E, \ell_2^d)$.
5. **No circularity.** The only result imported from the source paper is its already proved whole-space Euclidean case (Corollary 1.3), not the open-domain assertion under question.

This packet answers Question 1.5 only. It does not settle the universal dimension-free constant in Question 1.4 or complementability of $\mathcal{F}(\ell^2)$ in Question 1.6.

6 Novelty-search bounds and review recommendation

The run’s cheap indexes were searched for arXiv:1703.08384 and the core terms “open convex”, “Lipschitz-free”, “complemented in its bidual”, and “Banach–Mazur distance”. Bounded arXiv/web searches on 9 August 2026 used the exact Question 1.5 sentence, the exact source title, and close variants of those terms. They recovered the source paper and related later work, but no direct later proof of this statement. This is evidence only within the searched corpus, not a claim of exhaustive novelty.

Specialist review should focus on the functorial sandwich $RQ_H I^{**}$ and on whether a later publication already records this short retraction argument. Mathematically, the key observations are standard and all norm estimates are explicit.

References

- [1] M. Cúth, O. F. K. Kalenda, and P. Kaplický, *Finitely additive measures and complementability of Lipschitz-free spaces*, arXiv:1703.08384v2 (2018).
- [2] E. J. McShane, *Extension of range of functions*, Bulletin of the American Mathematical Society **40** (1934), 837–842.